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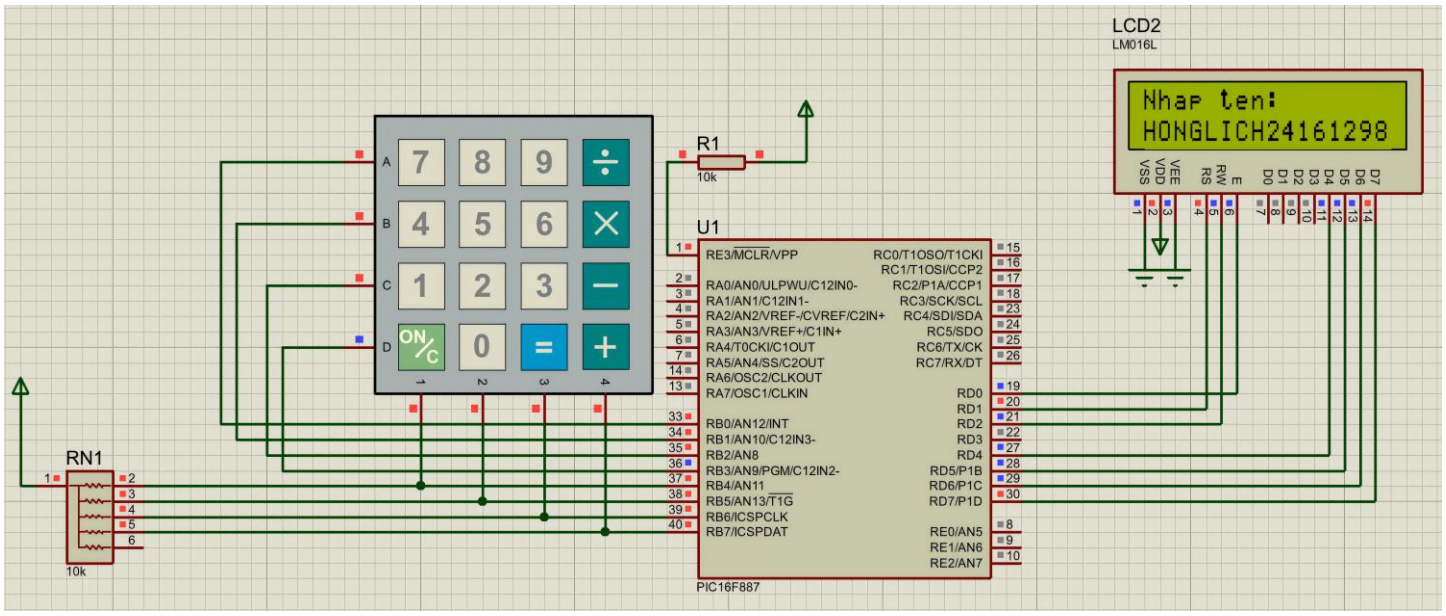
MSSV: 24161298

VXL Keypad LCD_extend




```
#include <16F887.h>
#fuses HS,NOWDT,NOPROTECT,NOLVP
#use delay(clock=8000000)
#byte ANSEL = 0x188
#byte ANSELH = 0x189
#include <lcd.c>
#define ROW1 PIN_B0
#define ROW2 PIN_B1
#define ROW3 PIN_B2
#define ROW4 PIN_B3
#define COL1 PIN_B4
#define COL2 PIN_B5
#define COL3 PIN_B6
#define COL4 PIN_B7
char keypad[4][4] = {
   {'H','O','N','G'},
   {'L','I','C','H'},
   {'2','4','1','6'},
   {'1','2','9','8'}
};
```

```
char get_key() {
  int row;
  for(row=0; row<4; row++) {
    output_high(ROW1); output_high(ROW2);
    output_high(ROW3); output_high(ROW4);
    switch(row) {
      case 0: output_low(ROW1); break;
      case 1: output_low(ROW2); break;
      case 2: output_low(ROW3); break;
      case 3: output_low(ROW4); break;
    }
    delay_us(20);
    if(!input(COL1)) return keypad[row][0];
    if(!input(COL2)) return keypad[row][1];
    if(!input(COL3)) return keypad[row][2];
    if(!input(COL4)) return keypad[row][3];
  }
  return 0;
}
void main() {
  int pos = 1;
  char last_key = 0;
  ANSEL = 0;
  ANSELH = 0;
  port_b_pullups(TRUE);
  lcd_init();
  lcd_putc("\fNhap ten:");
  lcd_gotoxy(1, 2);
  while(TRUE) {
    char key = get_key();
    if(key != 0 && key != last_key) {
      if(key == '*') {
        lcd_putc("\fNhap ten:");
```

```
        pos = 1;
        lcd_gotoxy(pos, 2);
      }
      else {
        if(pos > 16) {
          pos = 1;
          lcd_gotoxy(1, 2);
          lcd_putc("                ");
        }
        lcd_gotoxy(pos, 2);
        lcd_putc(key);
        pos++;
      }
    }
    last_key = key;
    delay_ms(10);
  }
}
```